LOT SYSTEMS CORPORATION

DOCUMENT: LOT-COMPUTER-HARDWARE-SPEC-v1

TITLE: COSMO® Cube v1.0 — The LOT Computer

CLASS: RESTRICTED // S-2 EYES

S-2: VADIK MARMELADOV

DATE: 2026-08-31

VERSION: 1.0 — PLAN, BOM POINTER, ROADMAP (PRE-HARDWARE, DESIGN LOCK PENDING)

00 // READING LOG — SOURCES THIS DOCUMENT IS BUILT ON

This document was commissioned directly by S-2 (Vadik Marmeladov) as a 19-point build brief: plan → BOM → roadmap for a physical hardware computer connected to lot-systems.com. Before writing a line of spec, the following were read in full:

docs/benchmark/LOT-MANIFEST.md

Line 31: "COSMO Hardware | brave-lamport-t9z5u8 | c7d353ef | 14/14 | BEST | 7 | +2610 | COSMO® Cube — complete hardware computer design v1.0." This is the FIRST reference to a COSMO® Cube hardware computer in the corpus — 2,610 lines across 7 files, marked BEST (ship candidate), never marked SHIPPED. The branch `brave-lamport-t9z5u8` does not exist in this repository (`LOT-Computer`) — it was cut in a different repo checkout and was never migrated here. Its content is UNRECOVERABLE from this working tree. This document does not reconstruct that lost branch; it re-opens the COSMO® Cube v1.0 line item from the task brief forward, in this repository, which is now the dedicated home for the hardware computer track (repo name: LOT-Computer).

docs/corporate/LOT-CUBIQ-QUANTUM-CUBE-v0.md

Line 49-56, its own reading log entry on LOT-MANIFEST.md: "CUBIQ™ is LOT®'s object: a notification body, not a computer... related by lineage (father/son, LOT®/COSMO®) and should share no naming collision going forward." CUBIQ (motion-only, no screen, no camera, LOT® brand) and COSMO® Cube (screen + camera + button, general-purpose computer, COSMO® brand) are DELIBERATELY DISTINCT hardware lines. This document is the COSMO® Cube line. It borrows CUBIQ's proven doctrine pieces where they generalize cleanly (table-as-power-surface, motion/light restraint, nano-ceramic/stainless material register) but does not merge the two products.

docs/corporate/LOT_ROBOTICS_COSMO.md
The COSMO® brand's founding document: named for Kuzya Cosmo Marmeladov, "a robot that carries the behavioral fingerprint of its owner." COSMO® Cube v1.0 is Phase 0 of that roadmap — a stationary, non-actuated computer, not yet the Phase 3 robot (2028-2029, $2,500-$5,000/unit) named in that document's revenue table. Building the Cube first de-risks the sensor stack, the LOT API connector, and the manufacturing partner relationship before COSMO® attempts anything that moves under its own power.

docs/technical/OS_API.md
The existing `/api/os/*` surface (status, version, insights, performance, diagnostics, config) is the CONSUMER-facing read side of a LOT profile. Section 05 of this document defines the WRITE side the Cube needs — a hardware-originated event reaching a user's Log — and proposes it as a new, clearly-marked extension rather than assuming it already exists.

docs/corporate/LOT-TERMINAL-M2M.md and docs/corporate/LOT-TERMINAL-SYNC.md
The existing machine-to-machine intake protocol (device_id, operator, sensors[], timestamp; WebSocket real-time / HTTPS batch fallback) is the correct shape for the Cube's own telemetry. Section 05 of this document reuses this protocol rather than inventing a new one.

docs/technical/LOT-NODE-0-RIG-SPEC.md
Precedent for how this corpus writes a from-scratch hardware BOM: named components, street-price ranges, a build-order sequence, and an explicit floor/serious/max cost table. LOT-COMPUTER-BOM-v1.md (this document's companion) follows that format.

docs/corporate/LOT-TERMINAL-VISION.md
"Complexity → Simplicity" and the S-2 operator progression. The Cube is the first LOT® object an S-2 operator can build from a kit and then plug into a full commercial profile — the hardware embodiment of `lot hardware init` → `lot systems connect`.

README.md (this repo)
LOT is self-care through proactive, context-aware AI; the Memory Engine "asks the right question at the right moment." The Cube is the

physical channel for that same proactivity, off the phone screen.

No prior document in this repository specifies a two-piece stainless enclosure, a camera, a screen, a dedicated wireless-charging plate, or a 100-unit pilot run for the COSMO® Cube. This document is that specification, v1.0.

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01 // WHAT v1.0 IS AND WHAT IT IS NOT

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v1.0 IS:

- A pager-class hardware computer: it receives one-way, AI-composed notifications from lot-systems.com ("Coffee time!") and displays them on a small screen. It is not a two-way chat device and does not run a general app store.
- A single physical object built from two CNC-machined stainless steel shells, one polished, one carrying the active face (camera, screen, button).
- Paired at all times with a companion object: a flat silver charging plate, 40mm x 40mm x 5mm, that is both the Cube's Qi-class wireless charger and its resting anchor — the same "table is the power surface" doctrine CUBIQ already established (LOT-CUBIQ-QUANTUM-CUBE-v0.md, Section 02).
- A LOT API client: every notification it shows, every button press it registers, and every sensor reading it takes either originates from, or is reported back to, a specific LOT user's profile.
- A 100-unit pilot manufacturing run, PCB and enclosure sourced through PCBWay, sized to prove the design before any larger commitment.

v1.0 IS NOT:

- COSMO® the robot (docs/corporate/LOT_ROBOTICS_COSMO.md). The Cube does not move, does not carry a "soul transfer," and requires no Benchmark Arbitrage® gate. It is Phase 0 infrastructure for that later product, not the product itself.
- CUBIQ (docs/corporate/LOT-CUBIQ-QUANTUM-CUBE-v0.md). The Cube has a screen and a camera; CUBIQ deliberately has neither. Do not merge the two specs or the two SKUs.
- A general-purpose smart-home camera or a security appliance. The camera's job (Section 03) is on-device presence/gesture sensing for

the notification and pairing flow, not continuous recording or cloud video streaming.

THE PRINCIPLE

Ship the smallest true pager first. A Cube that reliably shows one AI notification, logs one button press back to lot-systems.com, and survives a 100-unit build without a hardware revision is a complete v1.0. A Cube that also tries to stream video, run local inference models, or double as a COSMO® robot brain before the pager primitive is proven is not v1.0 — it is a mood board.

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02 // PHYSICAL FORM

BODY Two-part stainless steel (SUS304, brushed base / polished option) shell, CNC-machined, precision-fit with four M2 stainless screws through the base shell into PEM inserts in the top shell — no adhesive, so a unit is field-serviceable.

FOOTPRINT 45mm x 45mm x 18mm assembled (fits inside the 40mm x 40mm charging-plate footprint with a 2.5mm reveal on each side, so the plate is visually the Cube's base, not a separate object on the desk).

FACE A — POLISHED Mirror-polished stainless, no seams, no printing. This is the "resting" face — when the Cube is not actively notifying, this is the face left showing. Reprises CUBIQ's anti-feed thesis in metal: a polished surface asks nothing of the operator.

FACE B — ACTIVE Camera (Section 03), round display (Section 03), and the "Copy" button (Section 05), arranged on a matte-finish inset panel so the mirror face and the display face are never the same surface — a user physically rotates the Cube between "quiet" and "readable" states, a deliberate, tactile mode switch with no software toggle.

CHARGING PLATE 40mm x 40mm x 5mm flat silver plate (anodized aluminum, not stainless — aluminum is the correct material for an embedded Qi coil; stainless attenuates the induction field). Houses the Qi transmitter coil, a USB-C input, and a single

status LED at the plate's edge (pairing/charge
state only — same restraint rule as CUBIQ Section
02: light is secondary and utilitarian).
WEIGHT TARGET <95g Cube assembled, <60g plate.
INGRESS IP42 (splash-resistant, not submersible) — a desk
object, not an outdoor sensor node; Section 03's
weather sensor reads ambient room/near-window
conditions, it does not require outdoor placement.

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03 // ELECTRONICS AND SENSOR STACK

Every component below is an off-the-shelf, qualified part — no custom silicon, no bespoke sensor fab. This is a deliberate constraint: v1.0's job is to prove the LOT API connector and the pager notification loop, not to prove a new sensor. See LOT-COMPUTER-BOM-v1.md for exact part numbers, suppliers, and 100-unit pricing.

COMPUTE ESP32-S3 (dual-core, Wi-Fi + BLE 5, hardware JPEG/
camera interface, AI-instruction extensions for
on-device wake-word / gesture inference). Single
SoC — no separate applications processor. Keeps
BOM cost and firmware surface area small for v1.0.
DISPLAY 1.28" round IPS/AMOLED, 240x240, SPI — round to
match the Cube's mirror-face geometry visually
when both faces are viewed edge-on. Renders short
AI-composed text only (e.g. "Coffee time!"), never
a scrolling feed — this is a pager, not a phone.
CAMERA OV2640 (2MP, JPEG-capable) — used for on-device
presence detection (is anyone at the desk) and a
one-time QR pairing scan during setup. No
continuous video is stored or streamed off-device
in v1.0; presence is a boolean signal, not a frame.
WEATHER SENSOR BME280 (temperature, humidity, barometric
pressure) — feeds the same weather block already
shown on LOT public profiles (README.md, Public
Profile System) with a physically local reading
instead of a geocoded API call, and doubles as an
M2M-eligible intelligence source (docs/corporate/
LOT-TERMINAL-M2M.md Format 3, "environmental

monitoring").

AI-GRADE SENSORS Off-the-shelf, chosen for edge-AI compatibility, not for novelty:

- LD2410 mmWave presence/micro-motion sensor (through-desk-safe, no camera required for basic "someone is here" detection — the camera stays off unless presence is already confirmed by mmWave, a privacy-first sensor order)
- ICS-43434 MEMS microphone, single, local wake-word only (no continuous audio capture or upload — matches LOT-TERMINAL-M2M.md's "no personal health data in M2M protocol" rule, extended here to "no raw audio leaves the device")
- LIS2DH12 3-axis accelerometer (desk-bump / pickup detection, orientation-aware display wake)

WIRELESS CHARGING Qi receiver IC (BQ51013B-class) + coil in the Cube base shell; Qi transmitter IC (BQ500410A-class) + coil in the charging plate. USB-C PD input on the plate only — the Cube itself has no exposed charge port, consistent with the sealed, seamless two-shell design in Section 02.

BATTERY 400mAh Li-Po pouch cell, UL1642-certified, protection-circuit included — enough for multi-day standby between plate visits given the pager's low duty cycle (screen wakes only to show a notification or a button-press confirmation).

BUTTON Single tactile switch under the Face B inset, IP-rated membrane cap, silk-printed "COPY" (Section 05).

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04 // FIRMWARE / SOFTWARE SPLIT (SEPARATE DOCUMENTS)

Per the brief's explicit instruction ("Firmware documents," "Software to connect with firmware," "Separate documents"), the Cube's stack is documented as two independent manuals, not one combined technical doc:

docs/technical/LOT-COMPUTER-FIRMWARE-MANUAL-v1.md
What runs ON the ESP32-S3: drivers, power management, the notification-render loop, the button IRQ handler, OTA update mechanism, and the on-device session-compression buffer (Section 06).

docs/technical/LOT-COMPUTER-SOFTWARE-MANUAL-v1.md
What runs OFF the Cube: the phone/web pairing flow, the LOT API connector service, the notification composition path (which piece of lot-systems.com decides WHAT/WHEN to send — reusing the AI-driven physical product delivery model in docs/corporate/LOT-CUBIQ-OPERATOR.md Section 04), and the Log-tab write endpoint (Section 05).

This document (LOT-COMPUTER-HARDWARE-SPEC-v1.md) stays mechanical/electrical/manufacturing only and does not duplicate either manual.

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05 // LOT API CONNECTOR — THE PAGER LOOP AND THE "COPY" BUTTON

INBOUND (lot-systems.com → Cube) — THE PAGER NOTIFICATION
An AI-composed short message (the Memory Engine, QOS mode changes, or a scheduled Job — see README.md's Memory Engine and QOS sections) is queued server-side against a specific device_id, exactly as docs/corporate/LOT-TERMINAL-SYNC.md already queues procurement notifications to an S-2 operator. Delivery path: WebSocket (wss://sync.lot-systems.com/hardware/notify, real-time) with HTTPS long-poll fallback — the same real-time/batch hybrid mode already specified in LOT-TERMINAL-SYNC.md Section "Sync Modes." Example payload:

{
"device_id": "cosmo-cube-000042",
"type": "pager_notification",
"text": "Coffee time!",
"source": "memory_engine",
"ttl_seconds": 600
}

OUTBOUND (Cube → lot-systems.com) — THE "COPY" BUTTON
Pressing COPY does exactly one thing: it appends the Cube's current notification (if the screen is showing one) or a bare presence event (if not) as a new entry in the operator's LOT Log — the same Log surface `src/client/utils/logTriggers.ts` already parses for slash-

command triggers. This is a NEW endpoint, proposed here as an extension of the existing OS API surface (docs/technical/OS_API.md), not something already implemented in this repo:

POST /api/log/hardware-append [NEW — proposed]
Authorization: Bearer <device_token>

{
"device_id": "cosmo-cube-000042",
"event": "copy_button",
"payload_text": "Coffee time!",
"timestamp": "2026-08-31T14:02:11Z"
}

Response mirrors the M2M intake pattern already specified in LOT-TERMINAL-M2M.md:

{ "status": "accepted", "log_entry_id": "log_9f21ac" }

A pressed COPY button is a single physical action that becomes a single Log entry — no batching, no editing after the fact. This keeps the hardware's write surface as small and auditable as the LOT-NODE-0-RIG-SPEC.md transparency layer requires of any system acting on a user's behalf ("no silent writes").

DEVICE IDENTITY

Each Cube is provisioned once, at first plate-charge, with a device_token scoped to exactly one LOT user profile — mirroring the S-2 operator registration flow in LOT-TERMINAL-SYNC.md. A Cube with no bound profile shows only a single idle message ("Not yet paired — open lot-systems.com/hardware") and accepts no button input beyond re-pairing.

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06 // SESSION COMPRESSION DOCTRINE

"Compress the information in each session" (brief, item 8) is implemented at two layers, both required for v1.0:

ON-DEVICE (firmware layer — see firmware manual Section 04)

A "session" is one wake cycle: presence detected → notification shown or button pressed → screen sleeps. Rather than streaming every raw sensor sample, the firmware buffers one session into a single compact record (delta-encoded BME280/LD2410/LIS2DH12 readings, notification id, button state, timestamp) and flushes ONE record per session over the LOT API connector — not a continuous telemetry firehose. This is the same discipline named in docs/technical/ MEMORY-ENGINE-COMPRESSION-ARCHITECTURE.md for the software Memory Engine, applied to hardware: densify before you transmit.

SERVER-SIDE (software layer — see software manual Section 03)
Each compressed session record is itself folded into the user's Memory Story rather than stored as raw device telemetry — the Cube contributes to "what patterns emerge in your self-care" (README.md) the same way a journal entry or mood check-in does, not as a separate unreadable data lake.

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07 // MANUFACTURING — PCBWAY, 100-UNIT PILOT RUN

PCB FABRICATION 2-layer, 0.8mm FR-4, ENIG finish, via PCBWay (pcbway.com) — chosen per brief item 1. PCBWay quotes and assembles both the PCB and SMT population (turnkey) from a single BOM/Gerber upload, which keeps a 100-unit pilot to one vendor relationship instead of splitting fab and assembly.
ENCLOSURE Stainless shells CNC-machined by a metal-parts vendor (PCBWay also offers CNC machining as a service alongside PCB fab — evaluate as a single-vendor option before splitting to a dedicated metal shop). Polishing (Face A) and bead-blasting (Face B inset) are separate finishing passes after machining.
PILOT SIZE 100 units (brief item 13) — large enough to amortize the one-time tooling costs (stencil, fixture, CNC program) across a meaningful sample, small enough that a design flaw found in unit #40 does not sink a warehouse of stock.
DFM GATE No unit ships in the 100-run until: (a) five hand-built prototypes pass a 500-cycle button-press + wireless-charge-cycle test, (b) the two-shell tolerance stack is confirmed to close with a consistent, rattle-free fit across a 10-unit sample, (c) the LOT API connector round-trip

(notification in, COPY event out, Log entry
created) is verified end-to-end on real
lot-systems.com infrastructure, not a mock server.

COST POINTER Full component-level and 100-unit-run costing is
in LOT-COMPUTER-BOM-v1.md — this document does not
duplicate the BOM table.

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08 // ROADMAP — v1.0 → PILOT → SCALE

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v1.0 — THE PAGER (THIS DOCUMENT)

Two-shell stainless enclosure, ESP32-S3 + round display + camera +
weather/presence/motion sensors, Qi charging plate, LOT API
connector (inbound notification, outbound COPY-to-Log), on-device
session compression. Firmware and software manuals shipped as
separate documents. 100-unit pilot run through PCBWay.

GATE: 100/100 units pass the Section 07 DFM gate; 30-day field test
across a subset of Usership-tier LOT accounts with zero un-paired
bricking incidents and zero lost COPY events.

v1.1 — MULTI-GESTURE NOTIFICATION LANGUAGE

Reuse CUBIQ's four-gesture vocabulary concept (LOT-CUBIQ-QUANTUM-
CUBE-v0.md Section 04) as a LIGHT/DISPLAY vocabulary instead of a
motion vocabulary — the Cube does not move, but the display can
differentiate "urgent" vs "ambient" notifications the way CUBIQ
differentiates NUDGE vs LEAP. Requires no new hardware.

v1.2 — LOCAL WAKE-WORD + ON-DEVICE GESTURE INFERENCE

Exercises the ESP32-S3's AI instruction extensions properly (v1.0
only uses the mmWave sensor for boolean presence). Still no cloud
audio/video — inference stays on-device, only the resulting label
(e.g. "wave detected") crosses the LOT API connector.

v2.0 — SCALE MANUFACTURING (RESEARCH TRACK, NOT A BUILD MILESTONE)

Named as the horizon, not committed here. Candidate directions:
injection-molded polymer variant (cost-reduced SKU alongside the
stainless flagship) and a second PCBWay run sized from actual v1.0
field-test defect data rather than an estimate. No gate criteria yet.

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09 // DOCUMENTATION SET (THIS BUILD CYCLE)

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docs/corporate/LOT-COMPUTER-HARDWARE-SPEC-v1.md this document docs/corporate/LOT-COMPUTER-BOM-v1.md components, links, 100-unit costing

docs/technical/LOT-COMPUTER-FIRMWARE-MANUAL-v1.md firmware (on-device)

docs/technical/LOT-COMPUTER-SOFTWARE-MANUAL-v1.md software (LOT API connector)

docs/corporate/LOT-COMPUTER-USER-MANUAL-v1.md end-user manual

docs/corporate/pdf/LOT-COMPUTER-USER-MANUAL-v1.pdf rendered PDF (brief item 7)

docs/corporate/pdf/LOT-COMPUTER-HARDWARE-SPEC-v1.pdf rendered PDF (brief item 7)

docs/LOT-SR-20260831-01.md this session's report

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10 // BRAND

COSMO® Cube v1.0 The object (this document's subject)

LOT Computer This repository's working name for the product line

COSMO®† LOT® The combined mark — hardware carries the COSMO®

brand (per LOT_ROBOTICS_COSMO.md), the connector

and account are LOT®

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END LOT-COMPUTER-HARDWARE-SPEC-v1
